

Introduction to Data Engineering

Hadi Fadiallah

About Me

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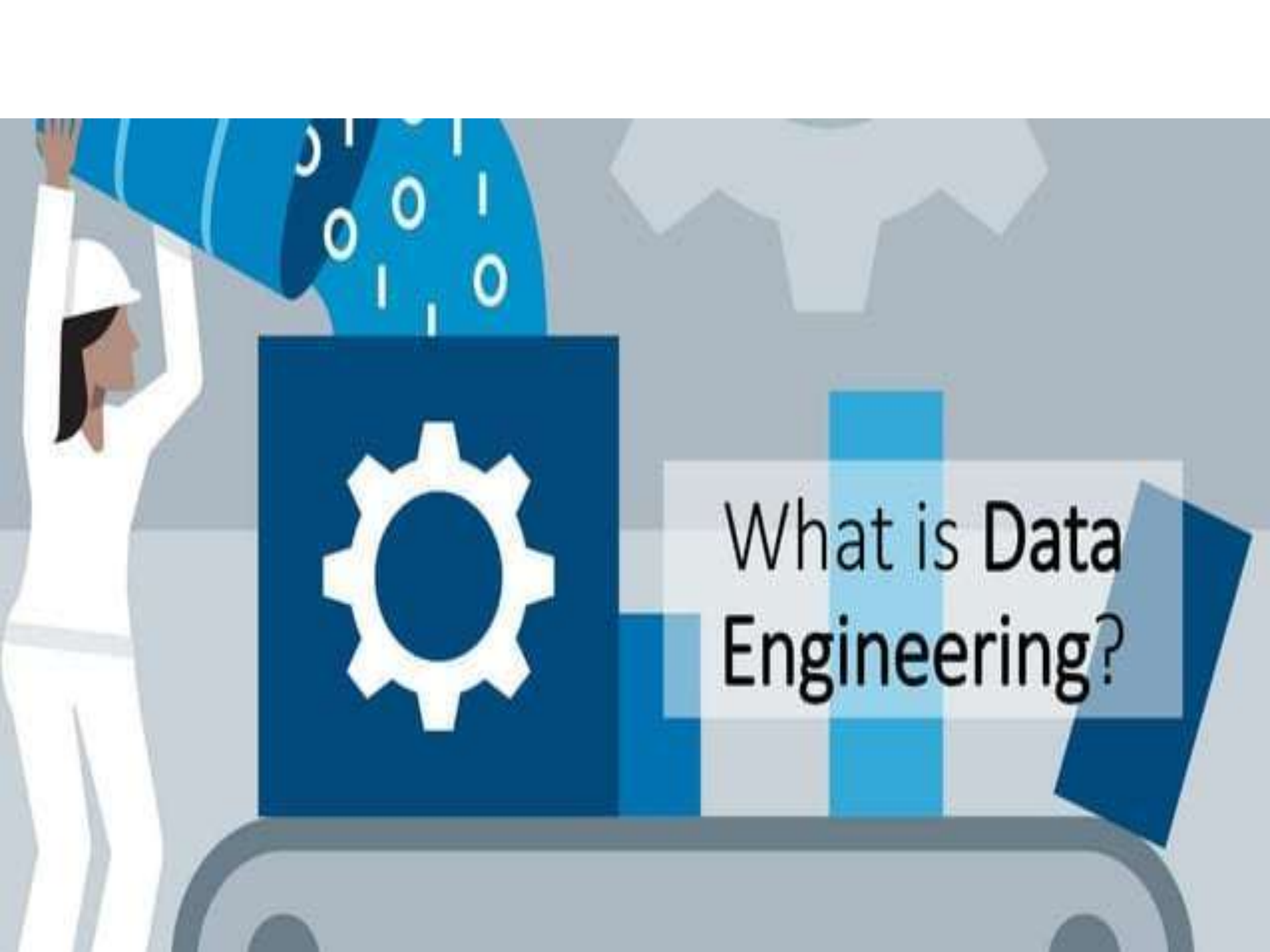
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Just a **Data Engineer!**



Plan

- What is Data Engineering?
- Data Engineer vs. Data Scientist vs. Data Analyst
- Understanding Data Management (Data Layers, DQS, MDS, Provenance)
- Distributed Computing
- Designing Data Pipelines (Choosing Paradigm / Technologies)
- Data Engineer Jobs / Required Skills
- Helpful Tips
- Online Courses



What is Data Engineering?

What is Data Engineering?

- Data Engineering is the act of:
 - Collecting data
 - Transforming (...) data
 - Validating data

“Making data consumable”

What is a Data Engineer?

AI/ML Engineer		BI Developer		Data Analyst	
Database Administrator		Report Developer		Data Developer	
Data Architect		Data Integration Specialist		ETL Developer	
		Data Scientist			

Data Engineer vs. Data Scientist

Data Scientist

also known as Data Managers, statisticians.



A data scientist will be able to take data science projects from end to end. They can help store large amounts of data, create predictive modelling processes and present the findings.

Skills: Mathematics, Programming, Communication



Will use programmes such as:
SQL, Python, R

Data Engineers

also known as database administrators and data architects.



They are versatile generalists who use computer science to help process large datasets. They typically focus on coding, cleaning up data sets, and implementing requests that come from data scientists.

Skills: Programming, Mathematics, Big data



Will use programmes such as:
Hadoop, NoSQL, and Python

Data Analysts

also known as business analysts.



They typically help people from across the company understand specific queries with charts.

Skills: Statistics, Communication, Business knowledge



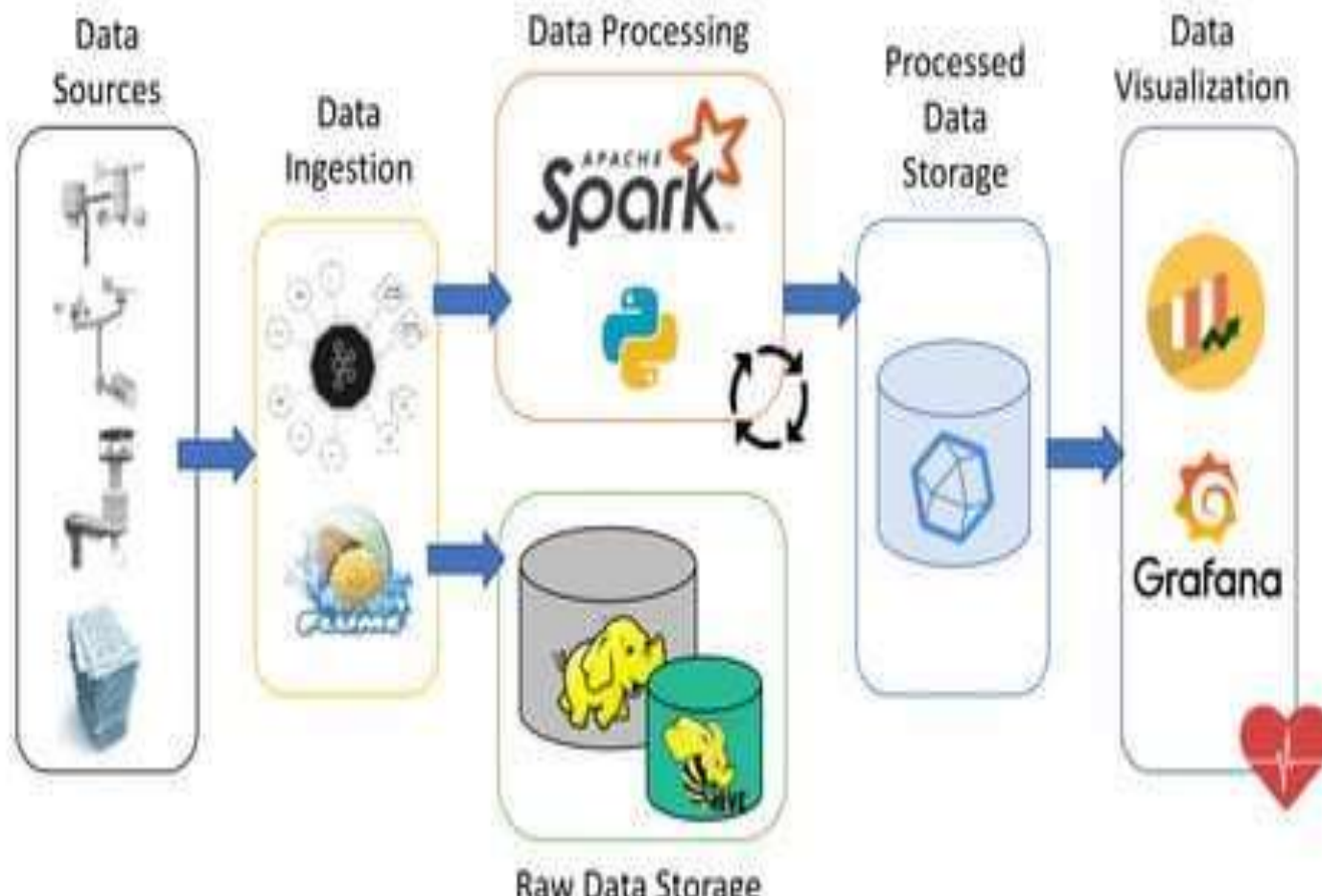
Will use programmes such as:
Excel, Tableau, SQL



DATA
MANAGEMENT



Understanding Data Layers



01 Data Discovery

Explore and discover data sources. Map data sources to correct and clean knowledge bases

01

02 Knowledge Management

Build and maintain a knowledge base of ways to clean and correct your data. Use this knowledge base to map data sources to clean data

02

Microsoft
SQL Server

**DATA QUALITY
SERVICES**
Process

04 Deduplication

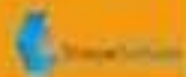
Match your data to your knowledge base and deduplicate data prior to use. Save yourself the trouble of data deduplication by automating this step

04

03 Correction & Standardization

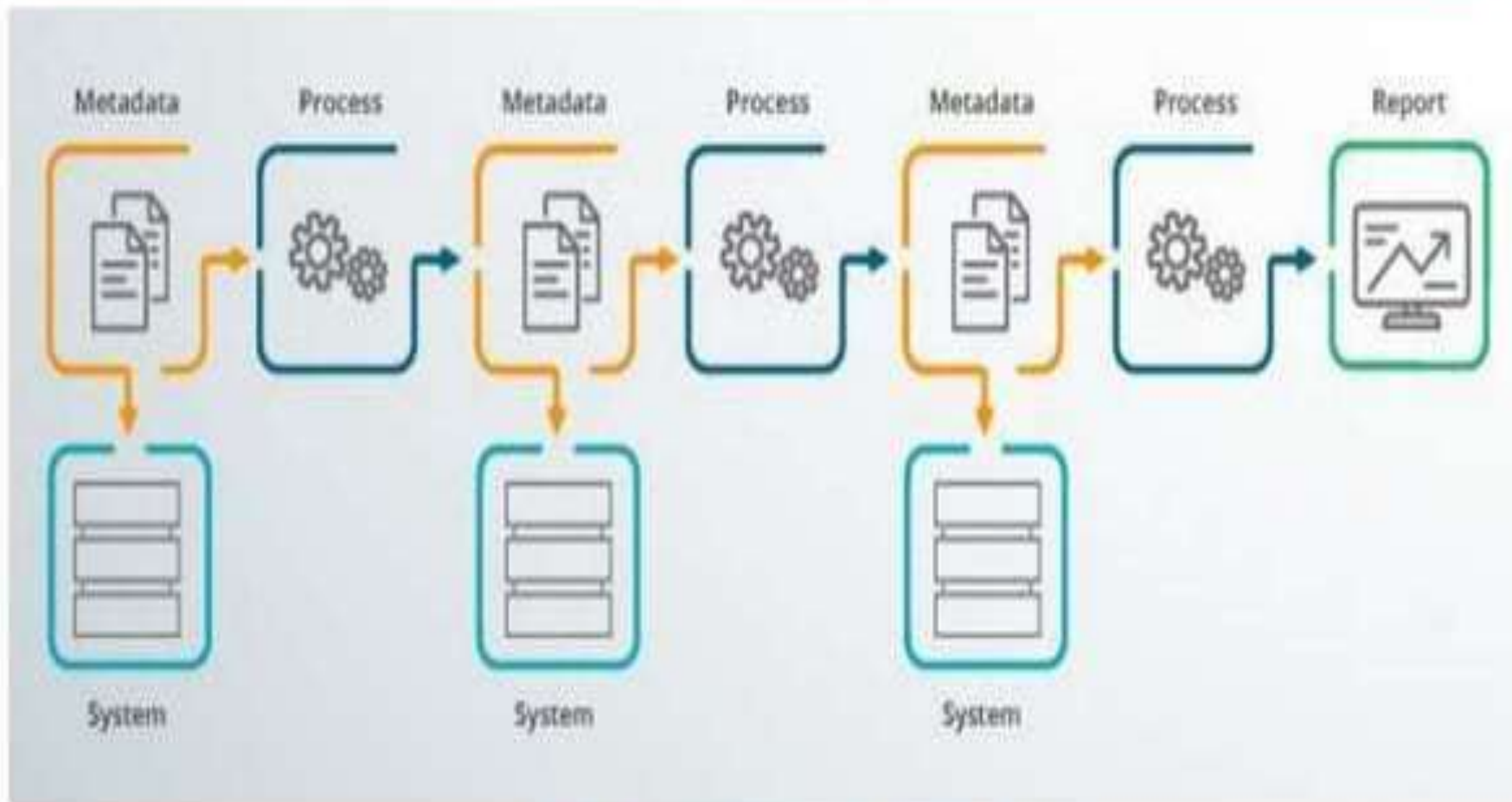
Correct and standardize your data to meet your data quality expectations. Output clean data in a standards-compliant manner to other systems

03

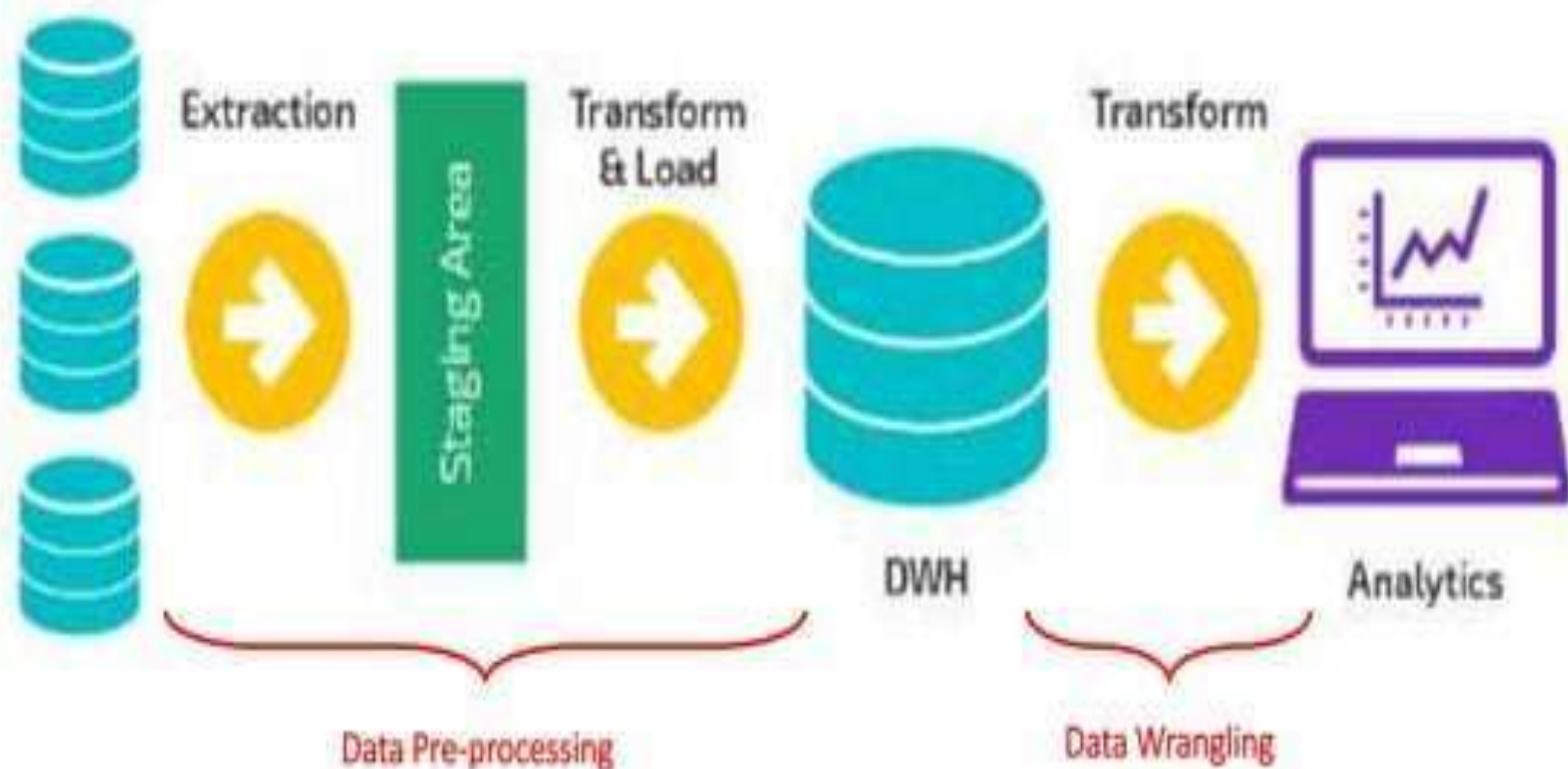


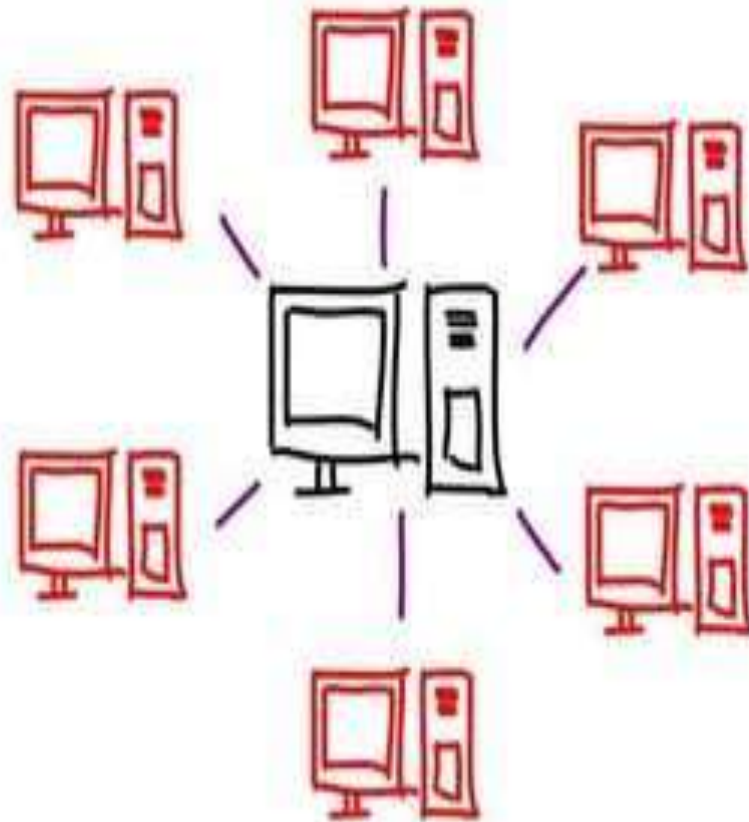


Data Provenance



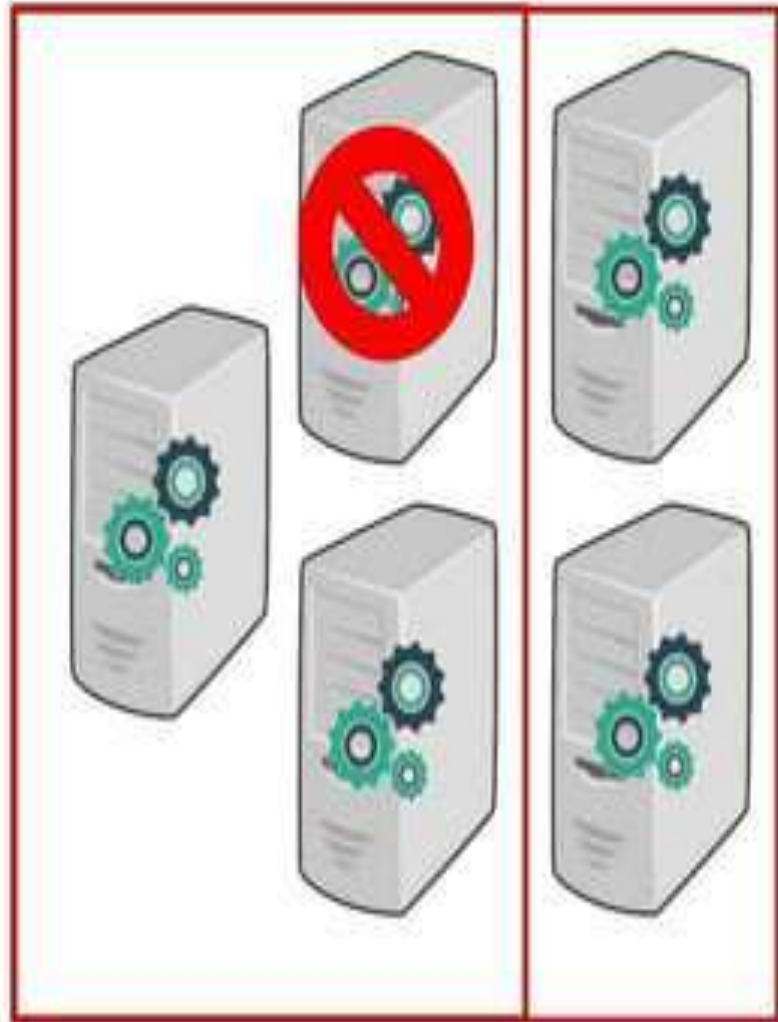
Data Wrangling vs. Data Pre-processing

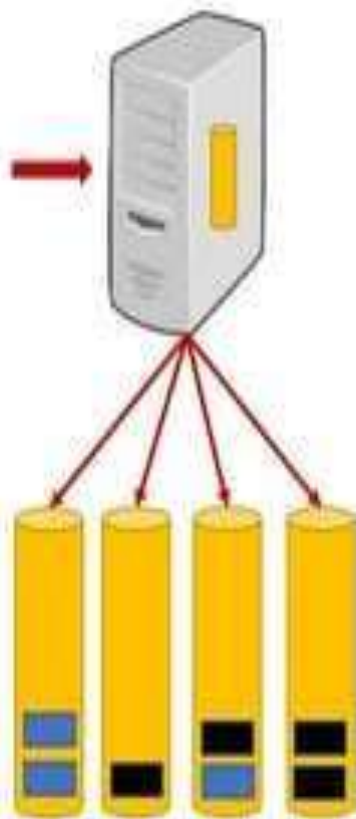




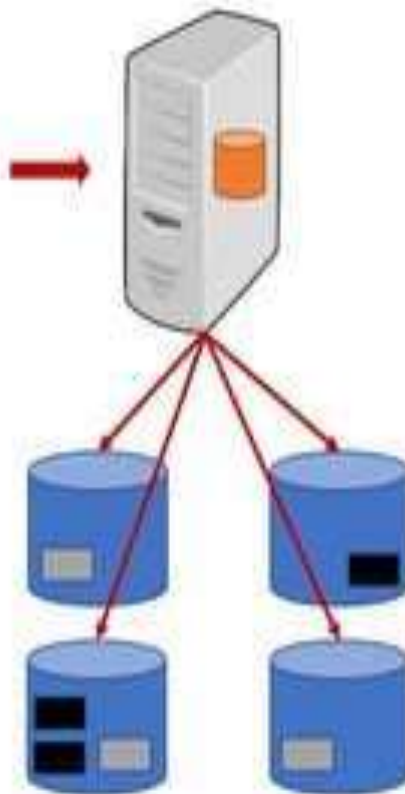
What are **Distributed** Systems?

- 1 Cluster = many machines (nodes)
- Parallel computing
- Scalability
- Fault-tolerance (High availability)





Data Ingestion



Data storage



Data Processing

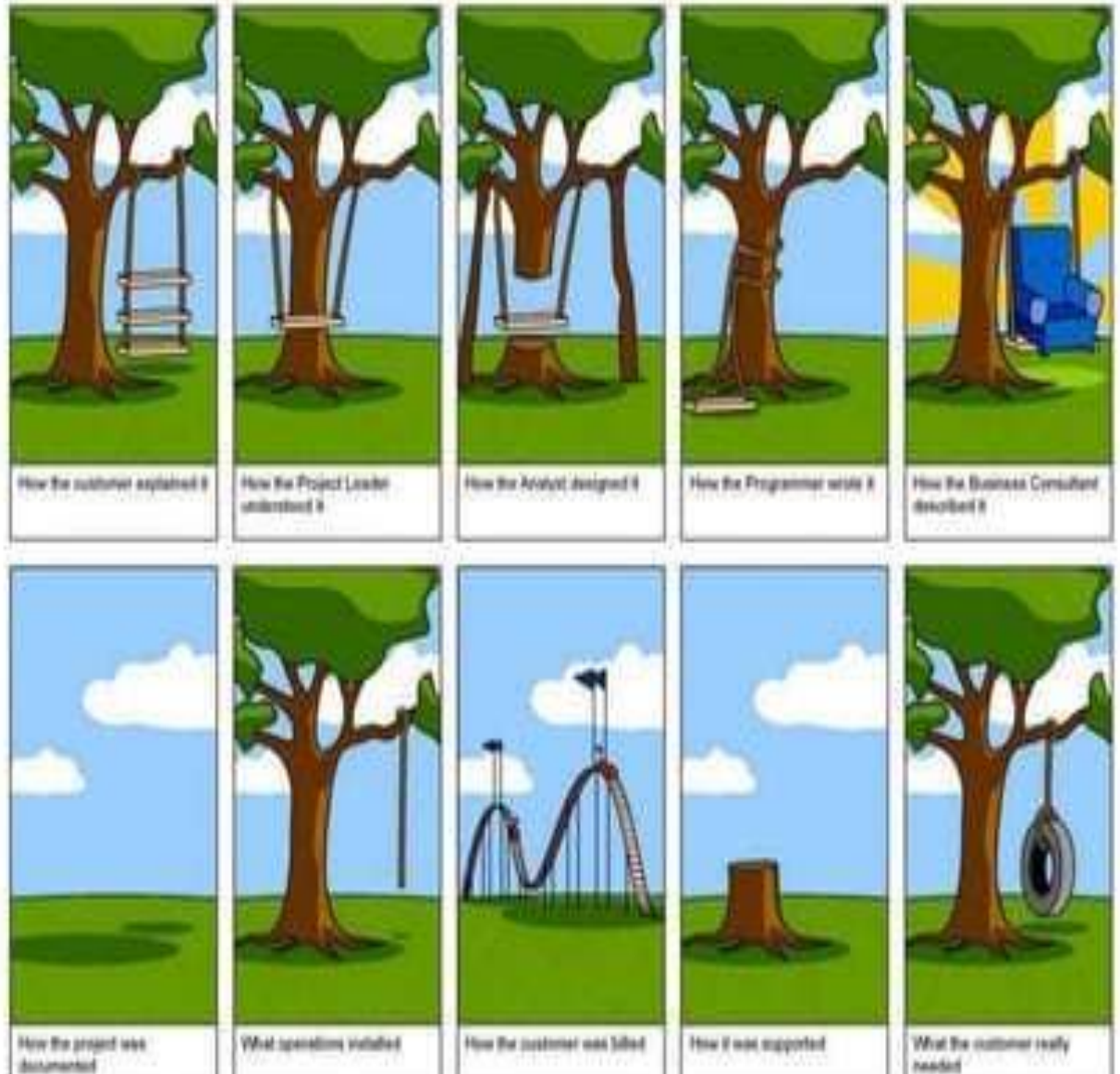


Designing Data Pipelines

Understanding Data & Requirements

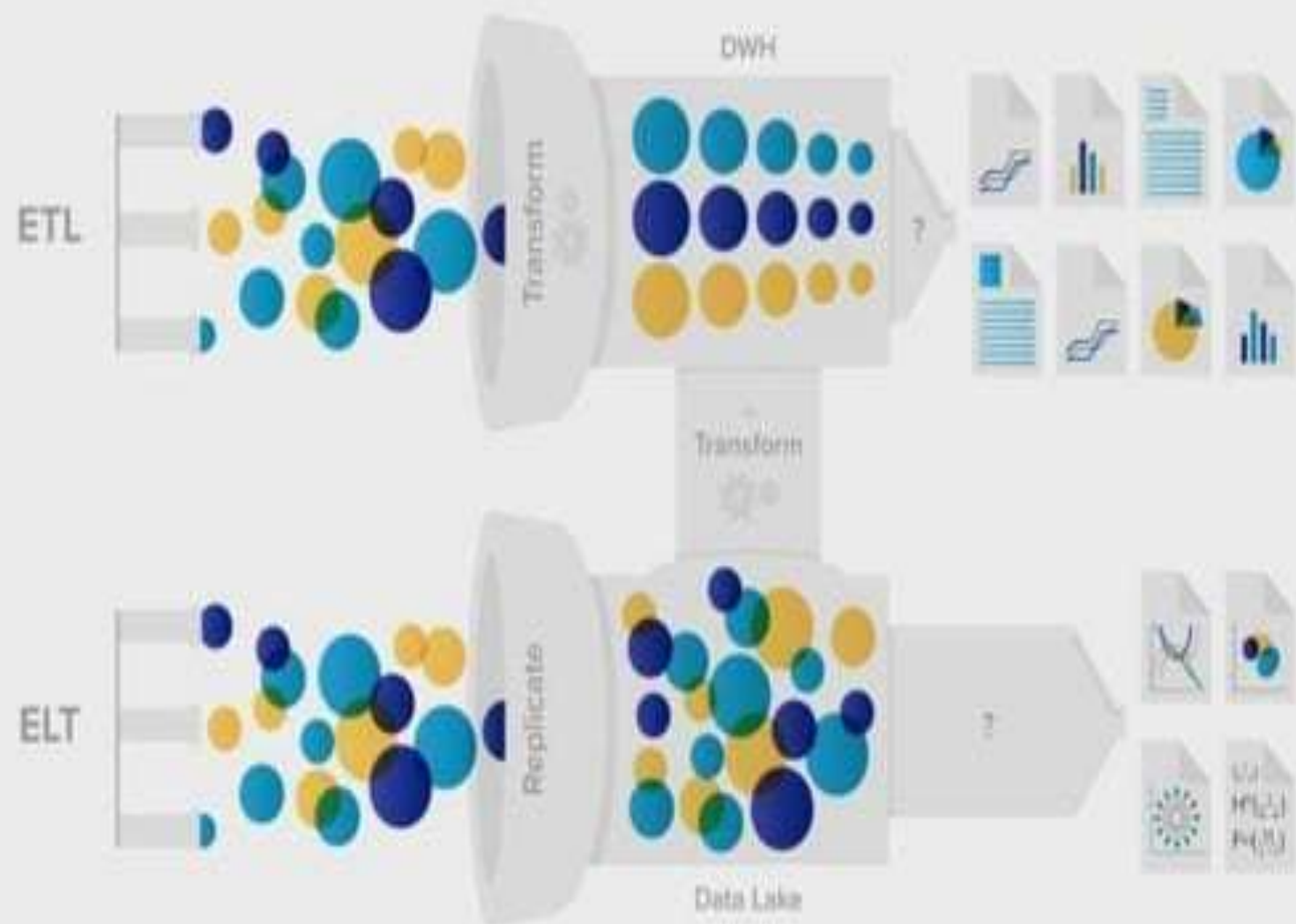
Source:

<https://firebrand.training/uk/learn/pmp/course-material/project-scope-management/collect-requirements>

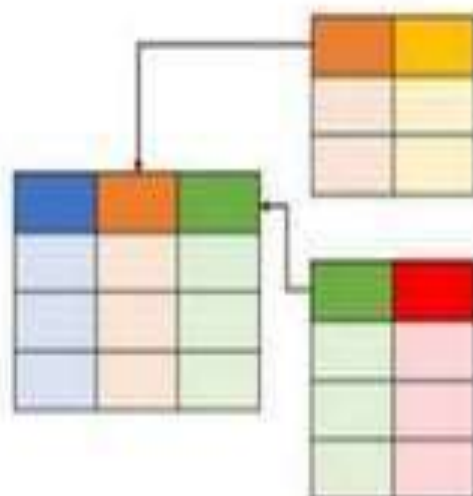


Choosing a Paradigm



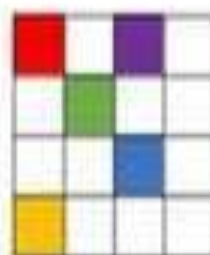


SQL DATABASES



Relational

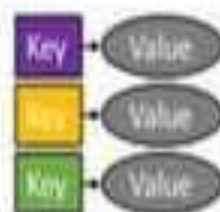
NoSQL DATABASES



Column



Graph



Key-Value



Document

DATA WAREHOUSE



- STRUCTURED FOR ANALYTICS
- CONSUMED BY PEOPLE AS A SELF-SERVICE
- FOCUSED ON DECISION MAKING

DATA LAKE



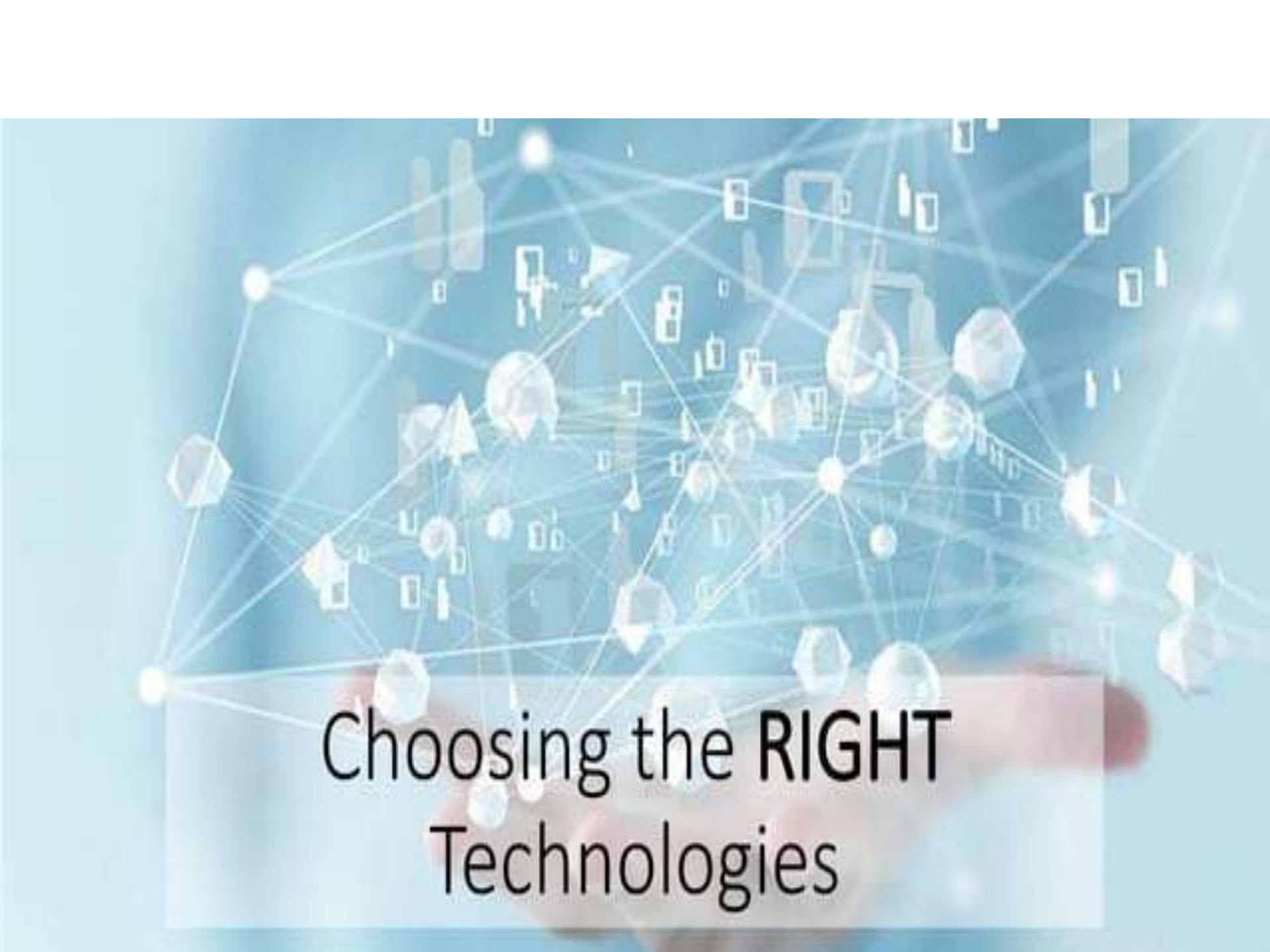
- (UN)STRUCTURED FOR DISCOVERY
- CONSUMED BY DATA PROFESSIONALS AND ALGORITHMS
- FOCUSED ON DEEP LEARNING, AI

DATA HUB



- STRUCTURED FOR DATA PORTABILITY
- CONSUMED BY PEOPLE AND APPS
- FOCUSED ON DATA INTEGRITY AND SPEED FOR SHARING

Source: <https://www.talend.com/resources/customer-360-data-hub/>

A hand is shown holding a transparent screen. On the screen, there is a complex network diagram with various nodes (circles, squares, hexagons) connected by lines. The background of the screen is a light blue gradient. The text "Choosing the RIGHT Technologies" is overlaid on the bottom half of the screen.

Choosing the **RIGHT**
Technologies

Relational databases

teradata.



Amazon RDS



ORACLE®

DATABASE



Microsoft®
SQL Server®



MySQL®



snowflake



PostgreSQL

SAP S/4 HANA

Data Integration



Azure Data Factory



Informatica



Microsoft
SQL Server
Integration Services



NoSQL databases



CouchDB
relax



Cassandra



mongoDB



membase



Big Data ecosystem





Data Engineer Jobs



Cloud Data Engineer (Azure)

Navy Federal Credit Union • Vienna, VA

azure python sql apache-spark etl

Yesterday • Easy apply • \$120k-215k • Remote



Data Engineer (Remote)

Blue Orange Digital • No office location

python amazon-web-services apache-spark

airflow

15d ago • \$30k-80k • Remote



Data Engineer

Platform.sh • No office location

etl sql python gcp amazon-web-services

9d ago • Remote



Sr. Data Engineer

Quadpay • No office location

etl python sql sysadmin

27d ago • \$140k-175k • Equity • Remote



Senior Data Engineer

Lynx Analytics • Singapore

python linux sql spark hadoop

26d ago • SGD 90k-120k • Equity • Limited remote • Visa sponsor



Data Engineer (m/f/d)

QuantCo • Berlin, Germany

python pandas sql-server data

23d ago • Limited remote • Paid relocation



Data Engineer

Connectera BV • Amsterdam, Netherlands

etl python sql

15d ago • 40k-55k • Visa sponsor



Data Engineer (Remote)

iFit • No office location

python sql r

4d ago • Remote



Senior Data Engineer

KdsLoop • Seoul, South Korea

amazon-web-services etl cloud

9d ago • Easy apply • Limited remote • Visa sponsor



Data Engineer

Legal & General • Cardiff, UK

etl

7d ago • £40k-45k • Limited remote



Python Data Engineer

Eagle Eye Staffing Solutions

Wroclaw, Dolnośląskie, Poland • Remote

You have a preferred skill badge

2 days ago • Easy Apply



Data Engineer

A Cloud Guru

Austin, Texas Metropolitan Area • Remote

\$115 - \$154k • Medical, Vision, Dental

Actively recruiting

3 days ago



Data Engineer Remote

Barnsley, GB • Remote

(TE)

Stack Overflow: <https://stackoverflow.com/jobs?q=data+engineer>

LinkedIn: <https://www.linkedin.com/jobs/search/?geoId=92000001&keywords=data%20engineer&location=Remote>

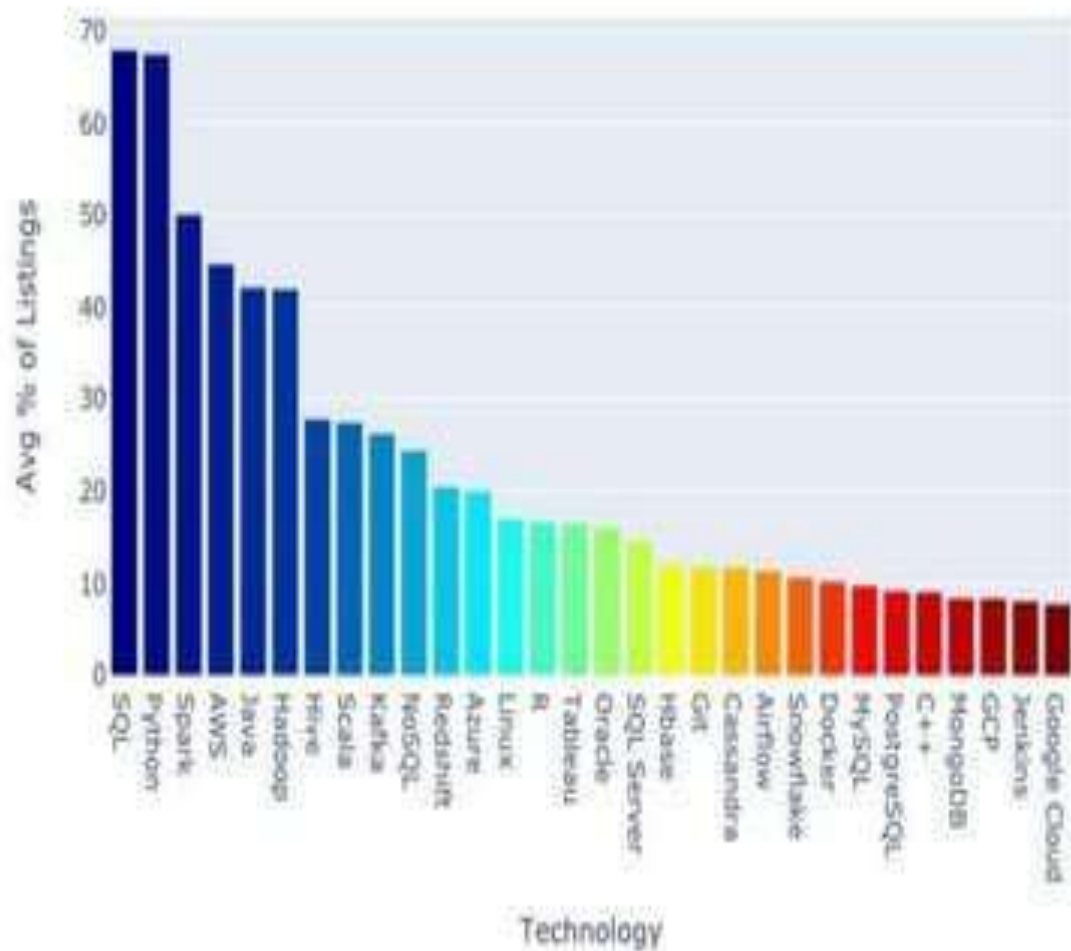
Required Skills

2. SQL, Python
3. Pandas library
4. AWS, Azure
5. Relational databases (SQL Server and MySQL)
6. Hadoop, Spark, Hive
7. NOSQL databases (MongoDB, Neo4j, and Cassandra)
- ...

1st Required Skill



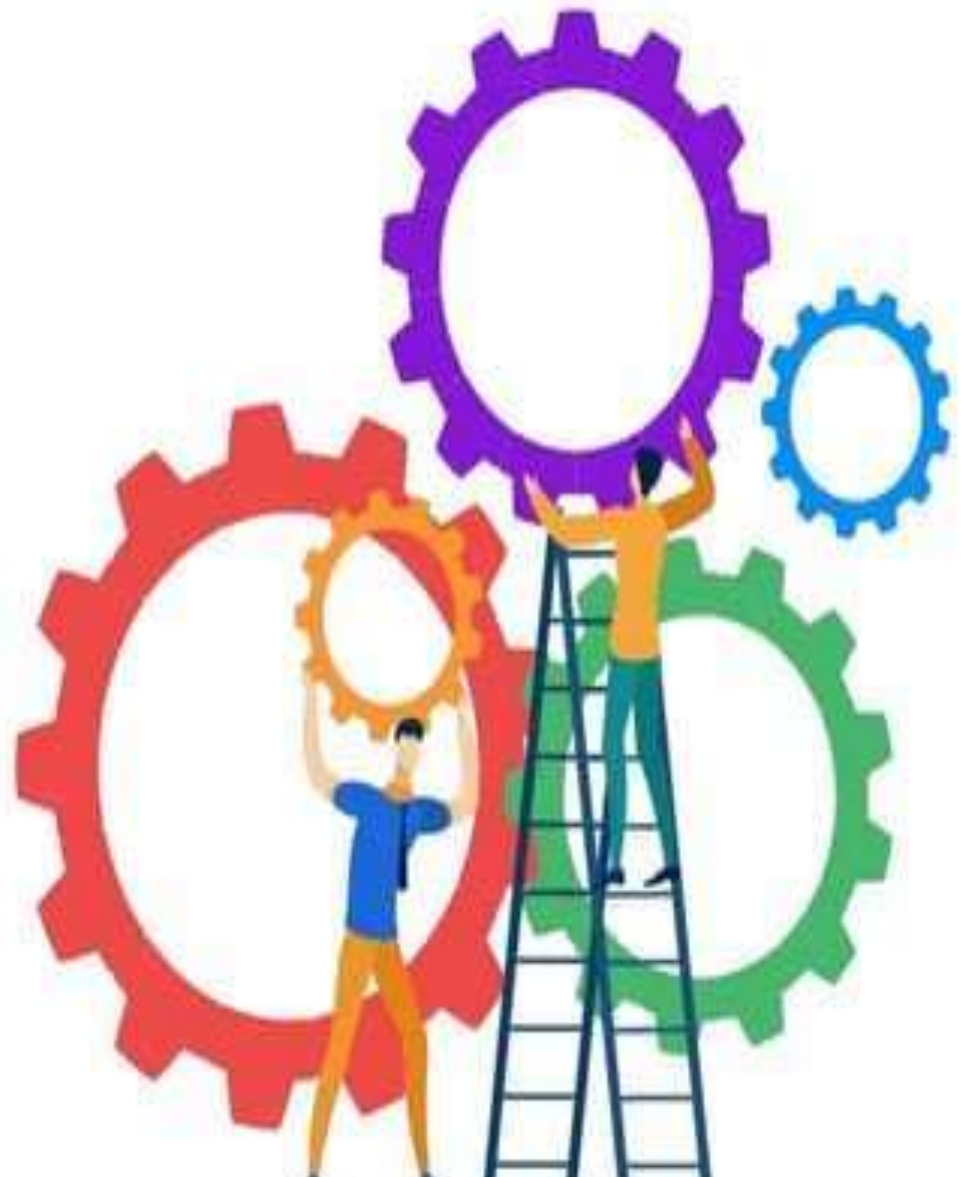
Technologies in Data Engineer Job Listings 2020





Helpful
Tips

*First, make it run
correctly (in an
acceptable way). Then
start improving
performance*



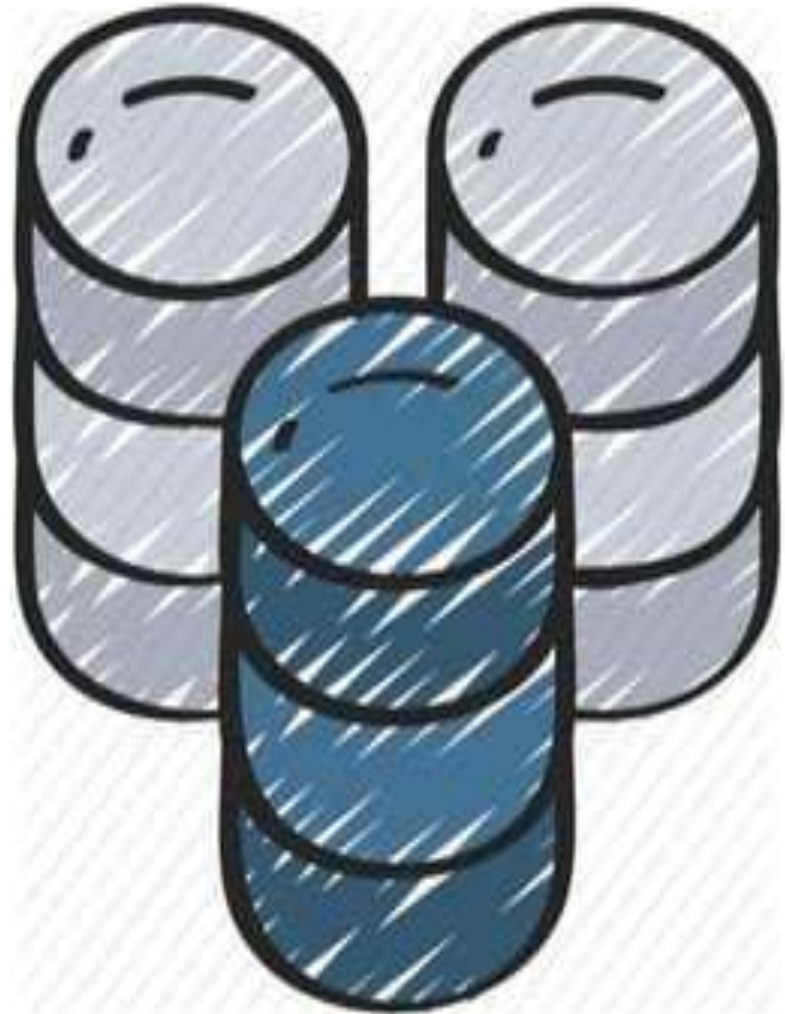
*Again, make sure
that you understand
well the data and
the requirements!*



Start with the basics (SQL, Python) and
DON'T
follow trendy words (Big Data, Big Data Engineer, Data
specialist..)

DON'T use Big Data technologies just because you have a huge volume of data.

Follow best practices



Online Courses

Udacity

Coursera



KHAN
ACADEMY



Udemy

USQ
UNIVERSITY OF
SOUTHERN QUEENSLAND

edX



- Coursera:

- [Google Cloud - Data Engineering, Big Data, and Machine Learning on GCP Specialization](#)
- [San Diego - Big Data Specialization](#)

- Udacity:

- [Data Engineering nanodegree](#)

- DataCamp:

- [Data Engineer with Python Track](#)

- IBM – CognitiveClass.ai

- [Free data science and data engineering courses](#)

- Udemy:

- [Data Science A-Z™: Real-Life Data Science Exercises Included](#)

Thank you



<https://hadifadl.github.io>